



**PRT661
DATA SCIENCE PRACTICE**

**ASSESSMENT 02
TECHNICAL DEVELOPMENT**

**FORECASTING ASSAULT RATES ACROSS NT
REGIONS TO SUPPORT PUBLIC SAFETY
RESOURCE PLANNING**

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SECTION 1 – Assessment 1 Report

1. Project overview and objectives

This project develops an end-to-end forecasting system for monthly, region level assault rates in the Northern Territory (NT), designed to strengthen operational planning within NT Police, Fire and Emergency Services (PFES). The work involves three core components:

- i. Building a secure and automated data pipeline covering acquisition, storage, and processing of NT crime datasets along with relevant datasets;
- ii. Developing a robust forecasting model capable of producing interpretable, region-specific monthly predictions; and
- iii. Delivering an interactive dashboard that presents historical trends, forecasts, and risk indicators for PFES operational use. The system ultimately supports PFES by improving resource allocation, enabling early identification of emerging hotspots, and enhancing evidence-based decision-making across the Territory.

The system supports PFES through better resource allocation, early hotspot identification, and evidence-based decision-making.

- Primary users: PFES operational planners and duty officers - shift commanders (patrol scheduling), regional operations managers (cross-station coordination), and strategic planning staff (quarterly staffing levels).
- Secondary users: PFES analysts needing historical trend data for budget/policy submissions.

The dashboard targets non-technical operational staff, prioritising clear visual summaries over raw statistical output.

2. Theme justification

The project's core theme is forecasting monthly assault rates across NT regions, within the crime analysis domain. Forecasting suits this data because it is genuinely time-series: assault counts show month-to-month dependence, seasonal patterns (dry vs wet seasons), and temporal autocorrelation - requiring models that capture temporal structure rather than static methods.

This theme aligns with the five mandatory data-science capabilities:

- Acquisition: pulling data from the NTG open data portal (Northern Territory Government, 2026)
- Storage: secure, structured retention of historical records.
- Processing: cleaning, aggregation, and feature engineering into monthly, region-level series.
- Analytics: the forecasting model itself, capturing trends, seasonality, and uncertainty.
- Visualisation: an operational dashboard showing patterns, predictions, and regional risk signals.

Together, these form a complete end-to-end system supporting PFES's operational planning and decision-making.

This project goes beyond training a single model - it spans the full data science lifecycle:

- Data sourcing & reconciliation: integrating 3 heterogeneous public datasets with differing formats and update cycles

- Storage & cataloguing: designing a secure, structured layer for historical and incoming data
- Data engineering: handling structural breaks in crime-recording and inconsistent regional boundaries
- Domain & ethical judgement: addressing Indigenous data governance and predictive-policing misuse risks to shape model outputs
- End-to-end delivery: a user-facing dashboard, not just a model artifact

The forecasting model is just one component of this pipeline. This system-level scope is what distinguishes the project as data science rather than applied ML.

3. Problem statement

PFES faces sustained resource pressure from high, volatile crime volumes across the NT. Monthly assault counts typically range 750 -1,100 incidents Territory-wide, with notable regional fluctuation (Northern Territory Government, 2026); demographic and alcohol-consumption factors may also contribute. These unpredictable surges strain PFES's finite staffing and patrol capacity (Northern Territory Police Association, 2025).

This volatility directly affects PFES's resource planning. Darwin, Alice Springs, Katherine, and Tennant Creek show distinct, shifting assault patterns, requiring frontline teams to adjust without advance warning. High domestic-violence involvement (often 60 - 70% of total assaults) and significant alcohol involvement further intensify demand on policing, investigations, and interagency coordination (Northern Territory Government, 2026). Without a mechanism to anticipate escalation, PFES remains reactive - reallocating resources only after spikes occur.

This project addresses that challenge with a forecasting system producing monthly, region-level assault predictions, backed by a secure data pipeline and dashboard. This gives PFES anticipatory visibility into hotspots and workload, enabling proactive staffing, patrol scheduling, and earlier interventions.

This aligns with PFES's intelligence-led approach, as seen in Operation Ventura (NTPFES, 2026), and extends it by helping PFES anticipate pressure points before they escalate - supporting more efficient resource use and sustained crime-reduction outcomes.

4. Initial architecture diagram

Table 1: Data sources

Datasets	Publisher	Date published	URL	File format	Rows	Columns
Population estimates for NTG Service Regions	Northern Territory Government	07 March 2024	https://data.nt.gov.au/dataset/nt-government-regions-1986-2025	XLSX	17,280	7

(1986 to 2025)						
NT Crime Statistics 2020-2023	Northern Territory Government	16 January 2024	NT Crime Statistics November 2023 - Dataset - NTG Open Data Portal	CSV	46037	10
NT Crime Statistics Latest	Northern Territory Government	11 June 2026	https://data.nt.gov.au/dataset/current-nt-crime-statistics-april-2026	CSV	9111	10
Wholesale Alcohol Supply by Quarter, 2023	Department of Tourism and Hospitality	13 November 2025	https://data.nt.gov.au/dataset/wholesale-alcohol-supply-by-quarter-2023	XLSX	29	12
Wholesale Alcohol Supply by Quarter, 2024			https://data.nt.gov.au/dataset/wholesale-alcohol-supply-by-quarter-2024		29	12
Wholesale Alcohol Supply by Quarter, 2025			https://data.nt.gov.au/dataset/wholesale-alcohol-supply-by-quarter-2025		29	12

The NT Crime Statistics dataset was selected because it provides monthly time-series crime data by region and offence type, making it suitable for temporal and spatial trend analysis. NTG population data were used to standardise crime rates per 100,000 population, enabling fair comparisons across regions with different population sizes.

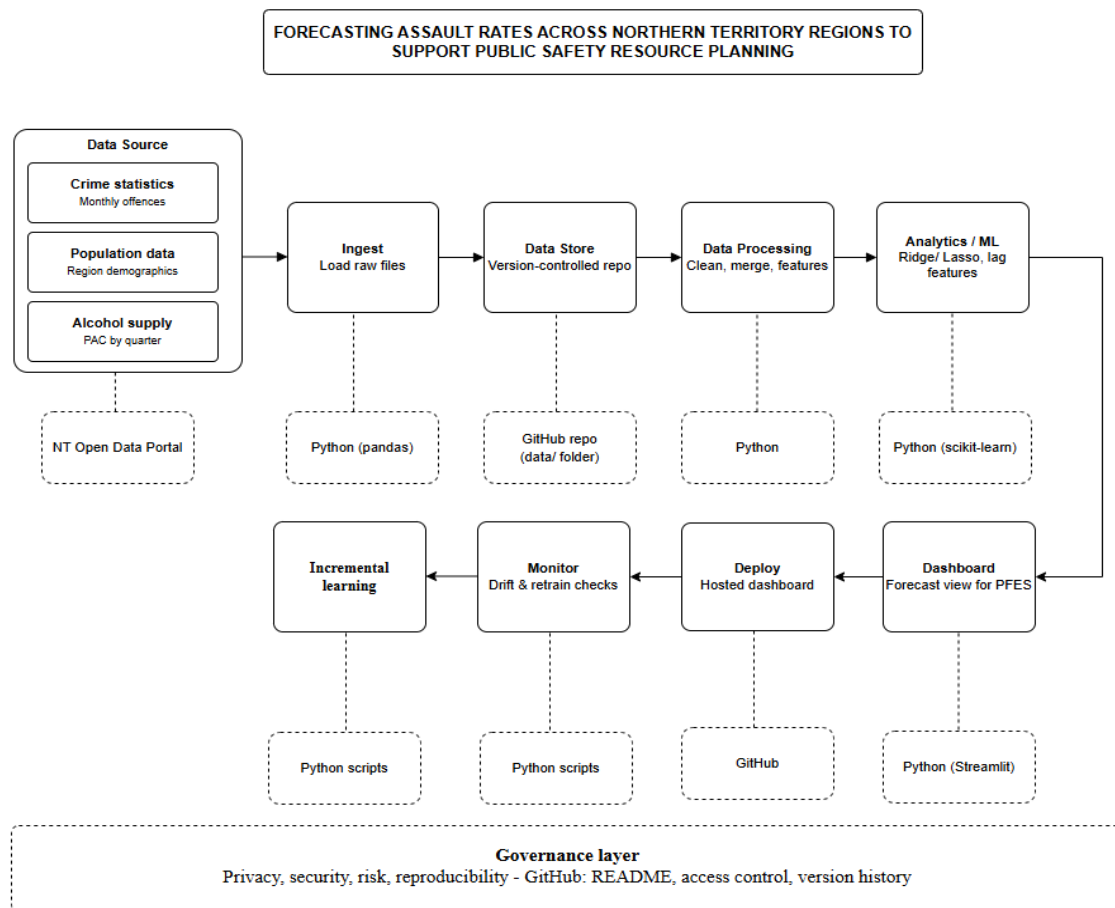


Figure 1: Project Architecture

Table 2: Pipeline stages, tools and purpose

Stage	Tool/ Technology	Purpose
Data Source	NT Open Data Portal	Three public datasets: crime statistics, population demographics, wholesale alcohol supply.
Ingest	Python (pandas)	Loads raw CSV/XLSX files from each source.
Data Store	GitHub repository	Version-controlled storage of raw and processed data, giving change history and access control in place of a cloud data lake.
Data Processing	Python (pandas)	Cleans, merges and engineers features (lagged indicators, cyclical month encoding, per-capita normalisation) across the three sources.
Analytics / ML	Python (scikit-learn)	Trains and validates Ridge and Lasso regression models to forecast monthly, region-level assault rates.
Dashboard	Python (Streamlit)	Presents forecasts and regional trends in an interactive view for PFES.

Deploy	Streamlit Community Cloud (via GitHub)	Automatically redeploys the live dashboard whenever changes are pushed to the GitHub repository.
Monitor	Python scripts	Checks forecast performance for drift and flags when the model should be retrained.
Incremental learning	Python scripts	Continuously adjusting the model parameters by training based on incoming data

Governance considerations spanning the full pipeline (privacy, security and reproducibility) are addressed through GitHub's built-in access controls, commit history and README documentation, and are detailed further in Section 8.

Although the architecture conceptually mirrors AWS's managed data pipeline pattern (S3 → Glue Data Catalog → Glue ETL → SageMaker → QuickSight, with SageMaker Model Monitor for post-deployment monitoring and SageMaker's built-in incremental training capability for scheduled retraining), the group project implements each stage using open-source, non-AWS tools, consistent with the module's guidance to rely on open-source tools for the group deliverable.

Technology roadmap. The current implementation uses open-source tools (pandas, scikit-learn, and Streamlit with GitHub providing version-controlled storage) to deliver a working proof of concept within the semester timeframe. Should PFES adopt the system operationally, the architecture is designed to migrate directly onto AWS-equivalent managed services with minimal redesign: the GitHub-based data store to Amazon S3, pandas-based data processing to AWS Glue ETL, the Analytics/ML stage to Amazon SageMaker for managed training and scheduled deployment, the Streamlit dashboard to Amazon QuickSight, the Monitor stage to Amazon SageMaker Model Monitor, and the Incremental learning stage to SageMaker's built-in incremental (continuous) training capability. Future enhancements under consideration include real-time or near-real-time data ingestion (replacing manual/batch downloads) and integration with additional predictors such as weather or event-calendar data to improve forecast accuracy.

5. Workflow plan

The workflow plan spans the full 12-week semester and is aligned to the four assessment milestones (Table 3). Weeks immediately preceding each submission are used for consolidation and reporting, other weeks are allocated to the team members responsible for the relevant pipeline stage, as shown in Table 4.

Table 3. Twelve-week workflow plan

Week	Focus	Deliverable	Owner
1	Team formation, theme confirmation, environment setup.	GitHub repository with README; Streamlit account set up; draft problem statement	All

2	Data collection and validation, architecture design	Architecture diagram; project management board populated; confirmed data sources	All
3	Proposal writing and initial risk analysis	Assessment 1 submitted: Project proposal and design	All
4	Load raw data into the repository; begin cleaning and harmonising six years of multi-source data; resolve region-mapping and data-regime discontinuities	Raw data in GitHub; cleaned, merged dataset in progress	Will (Ingest) Emma (Data processing)
5	Finalise ingest and storage structure in GitHub; complete data processing scripts; preliminary exploratory analysis	Working ingest/ store/ processing stages; EDA notebook	Will (Ingest, Data store) Emma (Data processing)
6	Consolidate pipeline evidence and baseline model into progress report	Assessment 2 submitted: Progress report and development	All
7	Feature engineering: lagged indicators, cyclical encoding, per-capita normalisation	Feature set finalised for modelling	Hendrick (Analytics/ML)
8	Model training and validation; begin dashboard build	Validated candidate model; dashboard prototype	Hendrick (Analytics/ML) Thomas (Dashboard)
9	Rehearse and deliver technical demonstration	Assessment 3 submitted: Group Technical demonstration and presentation	All
10	Incorporate feedback; finalise dashboard; deploy to Streamlit Community Cloud; implement monitoring and retraining logic	Deployed dashboard; monitoring stage implemented	Thomas (Dashboard, Deploy) Emma (Monitor)
11	End-to-end pipeline testing; draft final report sections	Full pipeline validated; report draft	All
12	Final review and submission	Assessment 4 submitted: Final Professional Report	All

6. Task allocation and team structure

Each team member owns one or two pipeline stages for the duration of the semester (Table 4), ensuring individually attributable contributions are visible in the GitHub commit history across all four assessments.

Table 4. Task allocation and team structure

Member	Primary role	Responsibilities
Will Nguyen	Ingest & Data Store	Loads the three raw datasets and maintains the version-controlled GitHub repository structure (raw and processed data folders)
Emma Tran	Data processing & Monitor	Cleans, merges and engineers features across the three data sources; implements post-deployment drift-checking scripts
Hendrick Dang	Analytics / ML	Develops, tunes and validates the Ridge/Lasso forecasting model
Thomas Tran	Dashboard & Deploy	Builds the Streamlit dashboard and manages its deployment and redeployment via Streamlit Community Cloud.

7. Initial risk analysis

The project has six main risks. These risks will be reviewed during each assessment milestone and updated when needed.

The first risk is changes in the crime data. The transition from PROMIS to SerPro in November 2023 and the ANZSOC classification transition in April 2025 may pose challenges in comparing old and new crime data. To minimise this risk, both transitions will be incorporated into the model.

The second one is that recent crime data may be incomplete because some offences are recorded later. To prevent incorrect results, the latest six months of data will not be included in the model's training and testing.

The third risk is limited data on alcohol supply. This data is only available quarterly and may not cover enough years. The team will collect older data where possible. However, the main forecasting model will still work without alcohol data.

The fourth risk is the same region may be named or bounded differently in different datasets. This can lead to wrong data matching. The team will work with one shared set of regions and validate the data after each merge.

The fifth is a delay in dashboard deployment. To reduce this risk, the team will create and deploy a simple version of the dashboard early on, then improve it throughout the project.

And the final risk is misuse or misinterpretation of the forecasts. If predictions are read as statements about particular communities, the system could cause harm rather than support safety. Small counts in low-population regions could also make incidents identifiable. Forecasts will therefore be shown only at region and month level, with uncertainty ranges, and small values will be suppressed.

8. Ethics, privacy and security considerations

Data privacy

The project only uses public and aggregated data from the Northern Territory Government and the Australian Bureau of Statistics. It does not use personal information or individual records. However, very small crime counts may still make some incidents easier to identify. Small values will therefore be hidden or grouped before being shown on the dashboard.

Indigenous data governance

Aboriginal people are strongly represented in NT crime statistics (Northern Territory Police Force, 2025), so the data must be used carefully and respectfully. The project follows the CARE Principles (Carroll et al., 2020) and the AIATSIS Code of Ethics (AIATSIS, 2020). The model will only make predictions at the regional and monthly level and will not make predictions based on Indigenous status.

Responsible use

The system is designed to help PFES plan staff and resources. It is not designed to predict whether an individual will commit a crime or to target specific addresses or communities.

Forecasts may not always be correct, so the dashboard will show uncertainty and explain the limitations of the data and model. This helps users avoid treating the predictions as exact results.

Security and reproducibility

Only public data and project code will be stored in the GitHub repository. Passwords, API keys and private information will not be uploaded.

The project will also keep its data-processing and analysis steps in version-controlled code. This allows others to understand, replicate, and verify the results.

GitHub repository:

<https://github.com/HendrickDang/PRT661-DATA-SCIENCE-PRACTICE>

Project management board (JIRA):

<https://hendrickdang.atlassian.net/jira/software/projects/SCRUM/boards/1>

9. References

- Australian Bureau of Statistics. (2025, June). *National, state and territory population*. <https://www.abs.gov.au/statistics/people/population/national-state-and-territory-population/latest-release>.
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SECTION 2 – Project Update Summary

• Changes Since Previous Assessment

Area	Previous State	Current State	Justification
Modelling approach	Ridge and Laso only	Added XGBoost	Captures potential non-linear patterns in lag features
Modelling validation	Single train/test split comparison.	Added evaluation.py: naive baselines, leakage-free CV, rolling-origin backtest	Responds directly to A1 feedback on validation rigour
Pipeline structure	Separate ingest.py and data_processing.py scripts	Merged into single End to End pipeline.py	Simplifies execution to one entry point; easier to maintain a single dependency / environment setup
Ethics	General statements	Specific design constraints and working agreements documented	Responds to A1 feedback on over-policing risk and governance rigour

• Contribution Summary

Team Member	Contribution
Thi Xuan Thanh Tran (Emma)	Data processing stage: cleaning, standardising, and merging the three source datasets.
Ngoc Anh Nguyen (Will)	Ingest and Data Store stage: built the data pipeline, then verified it end-to-end and fixed several runtime and code-quality defects before handoff.
Van Hoi Dang (Hendrick)	Analytics and Machine Learning: forecast task specification, validation design and leakage controls; built evaluation.py with naive baselines, rolling-origin backtesting and prediction intervals; set drift thresholds and retraining rules.
Le Nhat Minh Tran (Thomas)	Review the pipeline, making suggestions (feature engineering) and implement changes when necessary. Added XGBoost model to capture non-linear patterns.

• Evidence Locations

Evidence	Repository Location
GitHub	https://github.com/HendrickDang/PRT661-DATA-SCIENCE-PRACTICE
Jira	https://hendrickdang.atlassian.net/jira/software/projects/SCRUM/boards/1/timeline?rangeMode=WEEKS
Final code to run	https://github.com/HendrickDang/PRT661-DATA-SCIENCE-PRACTICE/blob/main/Updated_End_to_End_pipeline.py
Evaluation code	https://github.com/HendrickDang/PRT661-DATA-SCIENCE-PRACTICE/blob/main/evaluation.py
Ingest and processing code	https://github.com/HendrickDang/PRT661-DATA-SCIENCE-PRACTICE/commits/main/End%20to%20End%20pipeline.py

SECTION 3 – Assessment 2 Technical Development

1. Data acquisition and storage

Three datasets were acquired (Table 1): NT Crime Statistics (2020-2023 and current), Population estimates (1986-2025), and Wholesale Alcohol Supply (2023-2025). Raw files are stored untouched in dataset/source/; any correction happens in code, never by hand-editing the source.

Acquisition and processing are both handled by End to End pipeline.py. The two stages were originally developed as separate scripts (ingest.py and data_processing.py) but have since been merged into a single pipeline file for easier maintenance and a single point of execution; the ingest logic now runs as Stage 1 of that script:

- Fuzzy filename matching: handles naming differences between teammates' downloads (punctuation, Windows "(2)" duplicate suffixes).
- Validation checks: empty dataframes, duplicated columns, fully-null columns flagged before proceeding.
- Provenance tracking: each run writes to dataset/raw/ plus a manifest.json recording row/column counts and warnings.

Storage follows a three-stage convention: dataset/source/ → dataset/raw/ → dataset/processed/, all produced by the corresponding stages within End to End pipeline.py. GitHub was chosen for version-controlled history and individually-attributable commits at no cost.

The Assessment 1 feedback correctly flagged GitHub as a weak primary data store for production use. A SQL layer (SQLite/PostgreSQL) was considered but not adopted this milestone, for three reasons:

- Scale doesn't warrant it: current volume (~72,000 rows across all sources) doesn't need the concurrent-access or indexing capabilities a relational database provides.
- Functionality already exists: the joins and aggregations a SQL layer would offer are already handled by the existing pipeline; adding one would duplicate rather than extend capability.
- Timing risk: refactoring the already-tested Ingest, Processing, and Engineering stages mid-semester could destabilise working code during the weeks reserved for modelling and dashboard delivery.

A structured store remains scoped in the technology roadmap (Table 2) and will be revisited if the dashboard (Weeks 7–10) requires concurrent read access beyond what flat files support.

5 Vs characterisation:

Table 5: 5Vs characterisation

5Vs	Characterisation
Volume	55,148 raw crime records; 17,280 population rows; 87 alcohol rows → 66 region-year population rows and one merged region-month panel after cleaning

Velocity	Crime: irregular, multi-month cycle. Population: annual. Alcohol: quarterly. No source supports real-time ingestion, batch refresh only
Variety	Two CSV transaction logs, one wide Excel demographic table, three quarterly Excel extracts, merged into one long-format panel
Veracity	PROMIS→SerPro recording change (Nov 2023), category-label mismatch between crime files, missing Nov 2023, alcohol data from 2023 only, mismatched region boundaries (7 police vs 6 NTG regions)
Value	Crime = forecasting target; population = per-capita normalisation; alcohol = candidate predictor (alcohol–violence link)

2. Data cleaning and preprocessing

Before any analysis can begin, the datasets need to be cleaned and standardised so they can be combined reliably. Three main problems had to be solved:

(1) The crime data switched to a new classification system mid-way through the study period.

(2) The geographic regions used in the crime files do not match those in the population files, and

(3) Certain flags in the crime records needed to be encoded correctly to remain consistent across both file formats.

Step 1: Crime Data Cleaning

The two crime files use different offence category labels. The table below lists all translations:

Table 6: Category translations

Old category (nt_crime_statistics_2020-2023.csv)	→	New category (nt_crime_statistics_latest.csv)
Acts intended to cause Injury	→	02 Assault
Homicide and related Offences	→	01 Homicide
Abduction - harassment and other offences against the person	→	04 Harm or endanger persons
Other dangerous or negligent acts endangering persons	→	04 Harm or endanger persons
Sexual assault and related offences	→	03 Sexual offences
Robbery, extortion and related offences	→	05 Robbery, blackmail, and extortion
House break-ins	→	061 Burglary – dwelling

Motor vehicle theft and related offences	→	07 Theft
Theft and related offences (other than MV)	→	07 Theft
Property Damage Offences	→	11 Property damage offences

November 2023 gap. November 2023 is absent from both crime files. It is retained as NaN and never imputed.

Binary encoding. ‘Alcohol involvement’ and ‘DV involvement’ flags are encoded as 0/1. The dash (-) is treated as 0. Critically, the flags are multiplied by ‘Number of offences’ rather than counting source rows:

```
for col, out in [("Alcohol involvement", "Alcohol_offences"),
                ("DV involvement", "DV_offences")]:
    flag = crime[col].map({"Yes": 1, "No": 0, "-": 0}).fillna(0).astype(int)
    crime[col] = flag
    crime[out] = flag * crime["Number of offences"]
```

Figure 2: Offence-weighted encoding of Alcohol and DV involvement flags

Step 2: Region Remapping

Crime data uses 7 NT Police reporting regions; population and alcohol data use the 6 NT Government (NTG) statistics regions. Using mismatched boundaries without correction would assign the entire Greater Darwin population to both Darwin and Palmerston separately, doubling the denominator and producing per-capita rates roughly half their true value. All crime rows are remapped in two stages.

- **Stage 1 – Direct mapping:**
 - Darwin + Palmerston → Greater Darwin
 - Alice Springs → Central Australia
 - Katherine → Big Rivers
 - Nhulunbuy → East Arnhem
 - Tennant Creek → Barkly
 - Unknown → Top End
- **Stage 2 – SA2-level mapping for NT Balance rows.**

NT Balance records carry no town identifier. Each row also has a Statistical Area 2 (SA2) field indicating the sub-region of the offence. A secondary lookup table (SA2_TO_REGION) maps 16 SA2 values to their parent NTG region, such as:

- SA2 = ‘Barkly’ → Barkly
- SA2 = ‘East Arnhem’ → East Arnhem
- SA2 = ‘Howard Springs’ → Greater Darwin
- Any unrecognised SA2 → Top End

Step 3: Population Data Cleaning

The population file has a 'Status' column ('Preliminary' / 'Final'). If a year had both statuses, summing across demographic breakdowns would double-count residents. The pipeline validates this before aggregation:

```
multi = pop.groupby("Year")["Status"].nunique()
if (multi > 1).any():
    raise ValueError(
        f"These years have more than one Status: {multi[multi > 1].index.tolist()}. "
        f"Pick one (prefer Final) before summing, or population doubles."
    )
pop = pop.drop(columns=["Status"], errors="ignore")

total = (pop.groupby(["Year", "Region"], as_index=False)["Population"].sum()
        .rename(columns={"Population": "Total_population"}))
```

Figure 3: Population Data Cleaning

Population summed across all demographic breakdowns for each Region x Year combination. Status column dropped after validation.

Data trimmed to the 2015-2025 analysis window → 66 rows (6 regions x 11 years).

3. Data engineering

Data engineering transforms the three cleaned source tables into a single analysis-ready panel. Five steps handle dataset merging, per-capita normalisation, lag feature construction, age-structure dimensionality reduction, and train/test partitioning.

3.1. Merging the Three Source Tables

Two left joins produce the merged panel. Crime rows carry Year, Quarter, and Region keys; alcohol joins on Year × Quarter × Region; population joins on Year × Region.

```
panel = crime.merge(alcohol, on=["Year", "Quarter", "Region"], how="left")
panel = panel.merge(population, on=["Year", "Region"], how="left")
```

Figure 4: Merging the Three Source Tables

PAC quarterly attachment. Each quarterly PAC value is attached to all three months in that quarter, not divided by three. PAC measures total supply available during the quarter; dividing would understate the supply signal and create artificial within-quarter variation.

PAC imputation (2023-2025 only). Years 2015-2022 remain NaN by design; backward extrapolation would introduce unsupported assumptions.

Table 7: Output file after merging

Property	Value
Output file	Nt_crime_merged_2015_2025.csv
Coverage	2015-2025, 6 regions
PAC null values	0 within 2023-2025 after imputation; 2015-2022 PAC NaN

3.2. Per-Capita Metrics

Raw offence counts and PAC volumes are heavily confounded with population size. Greater Darwin has the largest number of the NT population and will always dominate absolute counts. Two per-capita are derived to enable fair cross-region comparison.

Table 8: Per-Capita Metrics

Derived Feature	Formula	Unit	Purpose
Assault_rate_100k	$\text{Assault_offences} / \text{Total_population} \times 100,000$	Offences per 100k people	Comparable assault intensity across regions
Alcohol_per_capita	$\text{Total_PAC} / \text{Total_population}$	Litres PAC per person	Comparable alcohol supply across regions

3.3. Features Engineering

A series of new features have been constructed from existing features to capture a holistic view of the crime situation in the Northern Territory. The features and reasons behind them are demonstrated below

Log_assault_rate:

Since the crime rate is not normally distributed but rather follows a log-normal distribution (heavily right-skewed) due to seasonality characteristics. By transforming the assault rate logarithmically, we aim to stabilise the distribution, making it symmetrical which is essential for good performance of linear regression models (Limpert et al., 2001)

Sin_month and cos_month:

To preserve the cyclic behaviours of months, sine and cosine functions are applied to represent time since these trigonometric functions pursue cycles between “top” and “bottom” and have a settled cycle length which is perfect for modelling regular cycles such as months (Lindsey and Pavur, 2005)

Season:

This feature serves as an indicator for wet and dry seasons, which according to research by Blakeslee et al. (2018), have a statistically significant relationship with crime incidence.

Assault_rate_lag1, lag3, lag12:

Three lagged versions of the assault rate are constructed for each region — the value one, three, and twelve months prior - computed separately per region (groupby("Region")) so a lag never crosses into another region's history. The three horizons capture distinct temporal patterns: **lag1** captures short-term momentum (recent months tend to predict the next), **lag3** captures a quarterly trend that smooths month-to-month noise, and **lag12** captures the seasonal/yearly cycle, allowing the model to compare a month against the same month one year earlier rather than assuming a fixed sine/cosine seasonal shape alone. These lags form the temporal core of every feature variant tested in Section 4.3.

3.4. Train/Test Split

Because of the nature of time-series data, results are susceptible to previous information. The dataset needs to be split into train and test using a temporal cut-off, ensuring that training data only contains observations prior to the test period to avoid any leakage of future information (Hasanov et al., 2022).

```
train = panel[panel["Year"] <= 2022]
test = panel[panel["Year"] >= 2023]
```

Figure 5: Train/Test Split

4. Analytics and machine learning

4.1. Exploratory Data Analysis (EDA)

EDA serves as a critical stage following data ingestion and pre-processing, as it enables systematic inspection of the dataset for distributional characteristics, outliers, inconsistencies, and anomalous patterns that may affect downstream modelling (Komorowski et al., 2016). This diagnostic step ensures that data quality issues are identified early and addressed appropriately. Beyond quality assurance, EDA also plays a strategic role in shaping the feature engineering process. By revealing underlying relationships, trends, and variable interactions, EDA provides the empirical insight needed to design features that capture meaningful structure in the data and enhance model performance. As Mao (2015) emphasises, effective feature engineering is grounded in a thorough understanding of the data, making EDA an indispensable component of the modelling pipeline.

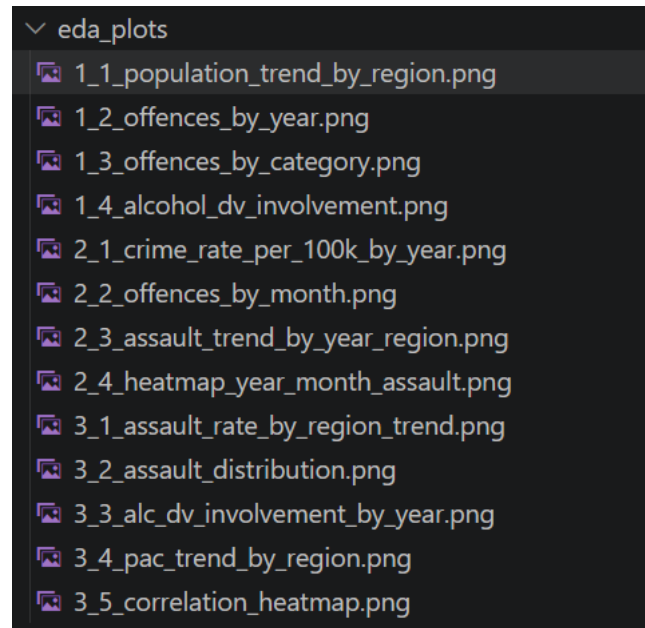


Figure 6: EDA plots

4.2. PCA of regional Age Structure

PCA was applied to 18 age-group population shares across 66 region-year observations as a diagnostic step, ahead of deciding whether age structure warrants inclusion as a model feature. PC1 explains 53.0% of variance; 5 components are needed for 90% (Table 9) - indicating the 18 age groups are highly correlated rather than independent signals.

Shares, not raw counts, were used deliberately: counts scale with region size and would leave PC1 at ~99%, merely re-measuring region size rather than age composition (Mishra et al., 2017).

This step is diagnostic only, components are not yet merged into the regression panel or included in FEATURES. Incorporating PC1-PC2 remains a candidate for future model iterations.

Table 9: PCA explained variance - age structure

PC	Explained (%)	Cumulative (%)
PC1	53	53
PC2	17.3	70.3
PC3	9.5	79.9
PC4	8.2	88.1
PC5	3.1	91.2
PC6	2.3	93.5

4.3. Feature Variant Selection

Four feature variants were compared using a baseline Linear Regression model with TimeSeriesSplit cross-validation over the training period (2015-2022, 504 rows after lag warm-up), leaving the 2023-2025 test period untouched. Each feature combination was standardised inside the cross-validation folds and trained under identical conditions, allowing a controlled comparison of predictive performance. RMSE was used as the primary selection criterion, supported by MAE, due to its sensitivity to larger errors and its suitability for the log-transformed assault-rate target.

Table 10: Feature variant comparison - Linear Regression, train 2015-2022

Variant	Features added	RMSE
V1: Temporal	sin/cos month, Season, lag1/3/12	0.2353
V2: + Region	V1 + region dummies	0.2146
V3: + Crime ctx	V1 + Alcohol_offences, DV_offences	0.2246
V4: Full (no PAC)	V1 + Alcohol_offences, DV_offences + region dummies	0.1794

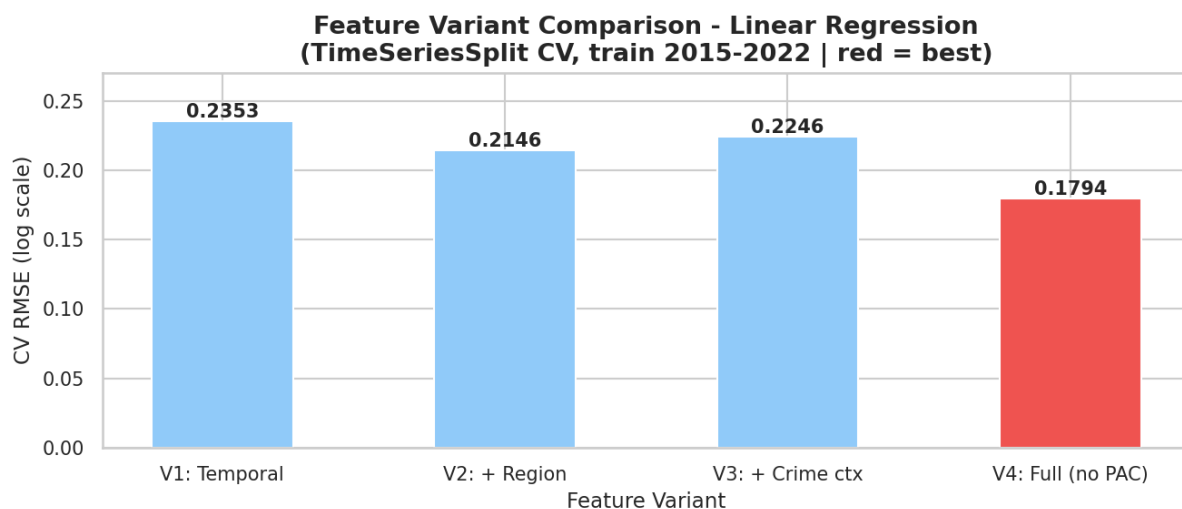


Figure 7: Feature Variant RMSE Comparison

V4 was selected as it achieved the lowest RMSE. Notably, V3 alone yielded a smaller gain than V2 - adding crime-context features without controlling for region-level baseline differences provides less independent signal, since regions differ systematically in both alcohol/DV offence volumes and assault rates. Only once region dummies are added alongside the crime-context features (V4) does the model benefit: the region terms absorb the between-region baseline differences, allowing the crime-context features to capture genuine within-region variation instead of confounding it with region identity. This is why V4 outperforms both V2 (region only) and V3 (crime context only) individually - the two feature groups are complementary rather than independently additive. Wholesale alcohol supply (PAC) was excluded from this comparison as it is only available from 2023 onward and would truncate the training window.

4.4. Multicollinearity Check and Hyperparameter Tuning

With an aim to maximise the model performance, a multicollinearity check was carried out by calculating the Variance Inflation Factor (VIF) of each feature and removing the ones with $VIF > 30$. High multicollinear features are known to have a negative effect on the regression model by hampering the interpretability of model predictors and resulting in highly volatile coefficient estimates (Dertli et al., 2024). In this study, while Alcohol_offences ($VIF = 24.1$), Reg_Barkly (17.2), DV_offences (16.6), and Reg_Top End (10.6) were flagged as elevated, all features remained below the removal threshold of 30, and no features were dropped.

Alcohol_offences	24.1	elevated - monitor
Reg_Barkly	17.2	elevated - monitor
DV_offences	16.6	elevated - monitor
Reg_Top End	10.6	elevated - monitor
Reg_East Arnhem	9.2	acceptable
Reg_Big Rivers	8.7	acceptable
Reg_Central Australia	7.3	acceptable
assault_rate_lag1	4.7	acceptable
assault_rate_lag12	4.4	acceptable
sin_month	2.0	acceptable
Season	2.0	acceptable

Figure 8: VIF scores for selected features

Another approach to optimising model performance is hyperparameter tuning. In this pipeline, tuning for linear models (Ridge and Lasso) is performed using alpha tuning with is to choose the most favourable alpha parameter that minimises the models' error in cross-validation (Deb, 2025).

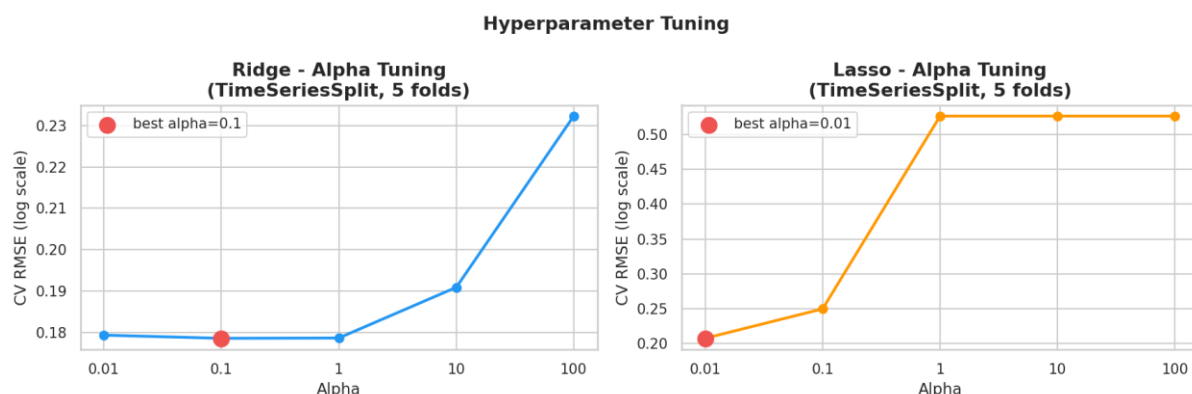


Figure 9: Ridge and Lasso CV RMSE across alpha values (TimeSeriesSplit)

4.5. Model Results and Evaluation

Table below compares four candidate models trained on Model A's feature set (Section 4.3-4.4), evaluated on the held-out 2023-2025 test period.

Table 11: Model performance comparison - Model A (no alcohol supply)

Model	RMSE(log)	MAE(log)	RMSE(/100k)	CV RMSE
Linear Regression	0.2010	0.1686	106.0	0.1795
Ridge ($\alpha=0.1$)	0.2004	0.1681	105.6	0.1786
Lasso ($\alpha=0.01$)	0.2153	0.1693	140.0	0.2071
XGBoost	0.1517	0.1227	84.5	0.1703

XGBoost performed best across every metric (RMSE 0.1517 log / 84.5 per 100k, lowest MAE and CV RMSE), indicating the predictors - particularly the lag features (Section 3.3) - contain non-linear structure a tree-based model captures better than a linear one. Residual normality check (Shapiro-Wilk $W=0.9755$, $p=0.0019$) rejected the normality assumption for the Linear Regression model, further supporting the selection of XGBoost as the final model. Paired t-tests on absolute error confirmed XGBoost significantly outperformed all three linear models ($p<0.001$), while differences between Linear Regression and Lasso were not statistically significant ($p=0.94$).

Among linear models, Ridge ($\alpha=0.1$) modestly beat plain Linear Regression, consistent with the correlated lag/region features flagged in the VIF check (Section 4.4). Lasso ($\alpha=0.01$) performed worst, likely discarding useful signal through coefficient shrinkage.

Per-region accuracy (Table 12) is best in Greater Darwin (RMSE 26.7, 7.5% error) and worst in Barkly and Top End (RMSE 131.3 and 114.6), with Top End under-predicted by 16.9% - suggesting the pooled model misses region-specific dynamics, a candidate improvement for future iterations.

Table 12: Per-region prediction accuracy (XGBoost, test 2023-2025)

Region	Actual (avg)	Predicted (avg)	RMSE	Error %
Greater Darwin	212.1	227.1	26.7	7.5
Central Australia	551.1	583.5	56.2	7.0
Big Rivers	624.0	603.4	83.9	-2.2
East Arnhem	275.1	256.9	39.7	-3.9
Barkly	852.5	770.6	131.3	-8.1
Top End	571.9	472.4	114.6	-16.9

As a secondary specification, Model B adds alcohol_per_capita (Section 3.2), trained on a much shorter window (2023-2024, test 2025 - 120 train rows vs Model A's 504) since alcohol data only starts in 2023.

Results are not directly comparable to Model A, the smaller sample and different test period mean RMSE differences reflect the data window as much as the model itself.

Figure 10 shows Model B's results. Unlike Model A, Linear Regression performs best here, with a small, consistent positive alcohol_per_capita coefficient (+0.0409) that shrinks toward zero under Lasso, suggesting alcohol supply is a weak, uncertain predictor once other variables are controlled for.

```

R10: Model B - with alcohol supply (train 2023-2024, test 2025) [A12] --
train 120 rows | test 72 rows

```

Model	RMSE(log)	MAE(log)	RMSE(/100k)
Linear Regression	0.1284	0.0838	57.1
Ridge (a=1)	0.1607	0.1139	79.8
Lasso (a=0.01)	0.2058	0.1553	116.1
XGBoost	0.1945	0.1407	94.1

```

standardised alcohol_per_capita coefficient:

```

Linear Regression	+0.0409
Ridge (a=1)	+0.1027
Lasso (a=0.01)	+0.0597
XGBoost	N/A (tree model)

Figure 10: Model Results and Evaluation

4.6. Forecast Evaluation and Benchmarking

This subsection documents the forecasting and evaluation work completed since Assessment 1. It responds directly to the Assessment 1 feedback, which asked the team to justify the choice of Ridge and Lasso, compare them against suitable baselines, and define the forecast horizon, validation approach, evaluation metrics, uncertainty calculation, leakage-control strategy, drift thresholds and retraining rules. Each is addressed below. The work is implemented in `evaluation.py` in the repository root, which runs independently of the main pipeline and writes its results to `Outputs/`.

Forecast task definition

Assessment 1 described the objective as monthly, region-level assault forecasting over a three-to-twelve month horizon but did not specify the prediction task precisely. It is now defined as follows.

- **Target:** assault offences per 100,000 residents, by NT Government service region, by calendar month. The model is fitted on the `log1p` transform of this rate and predictions are back-transformed before evaluation, so all reported errors are on the interpretable rate scale.
- **Granularity:** six regions (Barkly, Big Rivers, Central Australia, East Arnhem, Greater Darwin, Top End) across 116 months, giving 696 usable region-month observations after lag warm-up.
- **Horizons:** forecasts are produced and evaluated at 1, 3, 6 and 12 months ahead, spanning the operational range stated in Assessment 1.
- **Evaluation window:** the six most recent months are excluded because offences are still being recorded against them. This aligns the implementation with the commitment made in the Assessment 1 risk analysis.

Model selection and justification

Regularised linear regression is appropriate here for three reasons. First, the predictor set is dominated by lagged values of the target at one, three and twelve months, which are strongly correlated with one another; L2 regularisation is designed for exactly this collinearity, and Lasso additionally performs variable selection among competing lags. Second, the panel

contains 696 observations across six regions, which is small for high-variance learners and creates a substantial overfitting risk. Third, the end user is an operational planner rather than a data scientist, and linear coefficients can be read directly as the effect of a change in each predictor, supporting the interpretability requirement identified in Assessment 1.

Validation design and leakage control

Two sources of information leakage were identified in the existing pipeline and corrected. **First**, the feature-variant comparison selected the best variant using error measured on the 2023 to 2025 test set, and the final models were then scored on that same set. Reported performance was therefore optimistic, because the test data had already influenced a modelling decision. Variant selection now takes place on cross-validation folds drawn only from the training period. **Second**, the cross-validation routine fitted the feature scaler on the entire frame before splitting, so each fold was standardised using statistics that included later folds. Scaling is now performed inside a scikit-learn Pipeline, which refits the scaler on each fold's training rows alone.

Validation uses rolling-origin backtesting with an expanding window. At each origin the model is refitted on every month strictly earlier than that origin and then asked to predict the month h steps ahead. All predictors are lagged values or deterministic calendar terms, so no information from the target month or later enters the prediction. This measures forecasting performance rather than in-sample fit, and produces a distribution of errors across many origins rather than a single train-test split.

Baseline comparison

Assessment 1 compared candidate models only against one another, which cannot establish whether any of them add value. Two naive benchmarks were introduced: a seasonal naive forecast, which predicts the same month in the previous year, and a twelve-month moving average of the most recent complete year. Skill is reported as the percentage reduction in RMSE relative to the seasonal naive benchmark, so a negative value indicates a model performing worse than assuming last year repeats.

Table 13: Forecast accuracy by horizon, assault rate per 100,000 (rolling-origin backtest)

Model	Horizon (months)	MAE	RMSE	MAPE (%)	Skill (%)
XGBoost	1	66.4	90.7	14.4	29.7
Moving average (12m)	1	70.7	96.6	14.5	25.2
Ridge ($\alpha=0.1$)	1	75.1	99.4	16.2	23.0
Linear	1	75.1	99.4	16.2	23.0
Lasso ($\alpha=0.01$)	1	78.8	120.6	15.7	6.6
Seasonal naive (t-12)	1	91.7	129.1	19.2	0.0
Moving average (12m)	3	74.7	104.5	15.0	24.4
XGBoost	3	74.9	115.1	14.4	16.8
Ridge ($\alpha=0.1$)	3	88.5	123.1	18.2	11.0
Linear	3	88.6	123.2	18.2	10.9

Model	Horizon (months)	MAE	RMSE	MAPE (%)	Skill (%)
Lasso (a=0.01)	3	84.6	131.3	17.0	5.0
Seasonal naive (t-12)	3	95.6	138.3	19.5	0.0
Moving average (12m)	6	76.1	105.7	15.2	25.0
XGBoost	6	77.5	115.4	15.2	18.1
Ridge (a=0.1)	6	88.4	123.8	18.1	12.2
Linear	6	88.5	123.9	18.1	12.1
Lasso (a=0.01)	6	88.0	137.3	17.4	2.6
Seasonal naive (t-12)	6	97.4	141.0	19.7	0.0
Moving average (12m)	12	79.7	110.5	15.2	21.0
XGBoost	12	83.1	121.2	15.3	13.3
Ridge (a=0.1)	12	98.1	133.2	19.4	4.8
Linear	12	98.2	133.2	19.4	4.7
Seasonal naive (t-12)	12	96.2	139.8	19.2	0.0
Lasso (a=0.01)	12	97.8	153.2	18.8	-9.6

All six models evaluated on feature variant V2, selected on training-period cross-validation. Skill (%) is the RMSE reduction relative to the seasonal naive benchmark. Best model at each horizon shown in bold.

Interpretation of results

No single model dominates across the forecast range. XGBoost is the most accurate at a one-month horizon, achieving an RMSE of 90.7 against 96.6 for the twelve-month moving average, a skill score of 29.7 per cent against the seasonal naive benchmark. From three months onward the ranking reverses: the moving average leads at every remaining horizon, with an RMSE of 104.5 at three months against 115.1 for XGBoost, and the gap persists at six and twelve months.

The pattern is consistent across metrics and is not an artefact of the error measure chosen. XGBoost retains the lowest MAPE at one and three months while the moving average has the lower RMSE from three months on, which indicates that XGBoost makes smaller errors on typical months but larger errors on the extreme months that RMSE penalises more heavily. For a resource-planning application where large under-predictions are the costly failure mode, RMSE is the more relevant criterion and the moving average is the safer choice beyond the immediate month.

Both regularised linear models perform worse than the moving average at every horizon. Ridge reaches 23.0 per cent skill at one month but falls to 4.8 per cent at twelve, and Lasso performs worse than the seasonal naive benchmark at the twelve-month horizon, with a skill score of -9.6 per cent. This does not support the Assessment 1 plan, which proposed Ridge and Lasso as the primary models without benchmark comparison.

These figures are not directly comparable with the single-split comparison reported in Section 4.5. That comparison selects features and ranks models on one held-out period, whereas the results here come from repeated out-of-sample forecasts at a fixed horizon with leakage controls applied and provisional months excluded. Where the two disagree, the backtest is the more conservative estimate of operational performance and is the basis for the recommendation below.

The two stages also select different feature variants, and the reason is methodological rather than an inconsistency. The pipeline selects variant V4 on cross-validated RMSE computed on the log scale, which is the objective the models are fitted against. The evaluation module back-transforms predictions and computes the same statistic on the assault rate per 100,000, and on that scale V2 is preferred by a clear margin (85.1 against 107.6). The ranking is stable in both cases: V4 leads on the log scale and V2 leads on the rate scale, whether cross-validation is restricted to the training period or run over the full panel, so the divergence is driven by the choice of scale alone.

The two scales answer different questions. Log-scale error measures proportional accuracy, so Barkly at roughly 850 assaults per 100,000 and Greater Darwin at roughly 210 carry equal weight. Rate-scale error measures absolute accuracy, so the high-rate regions dominate. Both are legitimate, and the project retains both: V4 for the model comparison in Section 4.5, where log-scale fit is the relevant criterion, and V2 for the forecast evaluation reported here, on the principle that a model should be selected on the same metric the deliverable is reported in. The forecasts supplied to PFES are rates per 100,000, so rate-scale cross-validation is the appropriate basis for the operational recommendation.

The practical reading is that the short horizon and the planning horizon are different problems. At one month ahead there is enough recent signal for a non-linear model to exploit, and XGBoost does so. Beyond that, regional assault rates are dominated by a slow-moving local level, and additional model complexity contributes little. Three factors plausibly contribute to the weak medium-term signal: the panel is short relative to the number of parameters; the series contains two documented administrative breaks inside the last two years; and aggregating to six large regions smooths much of the local variation a richer model could exploit.

The recommendation for the operational system is therefore a split specification: XGBoost for one-month-ahead forecasts, and the twelve-month moving average as the reference forecast at three months and beyond. Any model proposed for the final system must demonstrate skill against the moving average at its intended horizon, not against the seasonal naive benchmark alone. This is a stricter standard than the one applied in Assessment 1.

Uncertainty quantification

Prediction intervals are constructed empirically from the distribution of backtest residuals rather than from a parametric assumption, since the residuals are not normally distributed. For each horizon the tenth and ninetieth percentiles of the residual distribution are added to the point forecast. Empirical coverage of the resulting intervals is between 78 and 80 per cent against a nominal 80 per cent, indicating the intervals are well calibrated. The intervals are wide, spanning roughly 230 to 246 assaults per 100,000, which is itself an important operational message: forecasts at this granularity carry substantial uncertainty and should be presented to PFES as planning ranges rather than point predictions.

Structural breaks

Binary indicators were added for the two documented changes in crime recording: the transition from PROMIS to SerPro from December 2023, and the ANZSOC reclassification from April 2025. Including these indicators did not change cross-validation error, because the cross-

validation period ends in 2022 and both indicators are therefore constant across every fold. They are retained because they are active in the evaluation period and because omitting a known level shift would bias the coefficients on the lag terms. This null result in cross-validation is reported explicitly rather than left implicit.

Drift monitoring and retraining

Forecast error is tracked as a rolling three-month MAE on the rate scale. A drift alert is raised when this rolling error exceeds the backtest MAE for the corresponding horizon by more than 50 per cent, which for the three-month horizon corresponds to a threshold of approximately 112 assaults per 100,000. Models are refitted monthly on the full available history when new crime data is published, rather than on a fixed schedule, so that each refit incorporates the most recent complete month.

Assessment 1 described an incremental learning stage, and the feedback correctly questioned whether the selected approach supports it. It does not: neither Ridge nor Lasso in scikit-learn implements `partial_fit`, so parameters cannot be updated online from streaming data. The stage is therefore clarified as scheduled full retraining on an expanded training window, not online parameter updating.

Limitations and next steps

Three limitations should be noted. The alcohol supply data covers only 2023 onward, so any specification including per-capita alcohol consumption is fitted on a much shorter window than the main models and its results are not directly comparable. The six-region aggregation was necessary to join the crime, population and alcohol sources onto a common geography, but it removes the sub-regional variation that would be most useful operationally. Finally, the evaluation covers 78 to 102 region-month predictions per horizon, sufficient to rank models but not to detect small differences between them with confidence.

Work planned for Assessment 3 includes testing whether a damped local-level model improves on the plain moving average, evaluating whether per-region models outperform the pooled specification, and integrating the prediction intervals into the dashboard so that forecast uncertainty is visible to end users.

5. Implementation progress

Since Assessment 1, progress has been made across five of the eight pipeline stages in Table 2:

- **Ingest & Data Store (Will) - *Complete*.** Six raw datasets ingested with validation and provenance tracking (Section 3.1).
- **Data Processing (Emma) - *Complete*.** Datasets cleaned, region-matched, and merged into a single analysis-ready panel (Section 3.2).
- **Data Engineering (Hendrick/Thomas) - *Complete*.** Per-capita normalisation, cyclical/lagged features, temporal train/test split (Section 3.3).
- **Analytics/ML (Hendrick/Thomas) - *Complete*.** Baseline pipeline covers EDA, PCA, feature-variant selection, multicollinearity screening, and four candidate models. A second, independent module (`evaluation.py`) adds naive baselines, leakage-free feature selection, rolling-origin backtesting, and empirical prediction intervals, responding directly to the Assessment 1 feedback on validation rigour (Section 4.6, Hendrick).

- Dashboard, Deploy, Monitor, Incremental-learning - *Not yet started*. Scheduled for Weeks 7–8 (dashboard) and Week 10 (deployment/monitoring) per the workflow plan (Table 3).

The updated workflow table below records progress against each week's planned deliverable at the time of this submission.

6. Updated Workflow

The workflow plan spans the full 12-week semester and is aligned to the four assessment milestones. Weeks immediately preceding each submission are used for consolidation and reporting, other weeks are allocated to the team members responsible for the relevant pipeline stage, as shown in table below:

Table 14. Twelve-week workflow plan

Week	Focus	Deliverable	Owner	Progress
1	Team formation, theme confirmation, environment setup.	GitHub repository with README; Streamlit account set up; draft problem statement	All	Done
2	Data collection and validation, architecture design	Architecture diagram; project management board populated; confirmed data sources	All	Done
3	Proposal writing and initial risk analysis	Assessment 1 submitted: Project proposal and design	All	Done
4	Load raw data into the repository; begin cleaning and harmonising six years of multi-source data; resolve region-mapping and data-regime discontinuities	Raw data in GitHub; cleaned, merged dataset in progress	Will (Ingest) Emma (Data processing)	Done
5	Finalise ingest and storage structure in GitHub; complete data processing scripts; preliminary exploratory analysis	Working ingest/store/processing stages; EDA notebook	Will (Ingest, Data store) Emma (Data processing)	Done

6	Feature engineering: lagged indicators, cyclical encoding, per-capita normalisation	Feature set finalised for modelling	Hendrick (Analytics/ML)	Done
7	Consolidate pipeline evidence and baseline model into progress report	Assessment 2 submitted: Progress report and development	All	Done
8	Model training and validation; begin dashboard build	Validated candidate model; dashboard prototype	Hendrick (Analytics/ML) Thomas (Dashboard)	Progressing
9	Rehearse and deliver technical demonstration	Assessment 3 submitted: Group Technical demonstration and presentation	All	
10	Incorporate feedback; finalise dashboard; deploy to Streamlit Community Cloud; implement monitoring and retraining logic	Deployed dashboard; monitoring stage implemented	Thomas (Dashboard, Deploy) Emma (Monitor)	
11	End-to-end pipeline testing; draft final report sections	Full pipeline validated; report draft	All	
12	Final review and submission	Assessment 4 submitted: Final Professional Report	All	

7. Project governance

Risk Management

Potential risks, including data quality issues, ethical concerns, and technical challenges, will be continuously monitored throughout the project. Mitigation strategies such as data validation procedures, adherence to ethical guidelines, and regular system testing will be implemented to minimise their impact. Risks will be reviewed periodically, and corrective actions will be taken promptly to ensure project objectives are achieved successfully.

Compliance

As this project involves the analysis and forecasting of crime-related data, strong governance and compliance measures are essential to ensure responsible data use and maintain public trust.

The project will adhere to Northern Territory Government data-handling policies (Northern Territory Government, 2024) to ensure that data is collected, stored, accessed, and managed securely throughout the project lifecycle.

The project continues to respect Indigenous Data Sovereignty principles by recognising the rights of Indigenous communities to govern the collection, ownership, and use of data relating to their people and communities.

In addition, all predictive analytics activities seek to comply with ethical standards to minimise bias, promote transparency, and support fair decision-making. Forecasting models will be regularly evaluated for accuracy, reliability, and potential unintended consequences.

To strengthen day-to-day governance, the team applies a set of working agreements throughout the project. All code changes go through a pull request reviewed by at least one other member before merging. Jira tickets move to "Done" only when the associated code has been merged and the relevant pipeline stage runs without errors. Blockers unresolved within two working days are escalated at the next team check-in. Any scope or dataset change is logged in the repository change log with a brief justification before work begins.

8. Risk analysis

The project has six main risks. These risks will be reviewed during each assessment milestone and updated when needed.

The first risk is changes in the crime data. The transition from PROMIS to SerPro in November 2023 and the ANZSOC classification transition in April 2025 may pose challenges in comparing old and new crime data. To minimise this risk, both transitions will be incorporated into the model.

The second one is that recent crime data may be incomplete because some offences are recorded later. To prevent incorrect results, the latest six months of data will not be included in the model's training and testing.

The third risk is limited data on alcohol supply. This data is only available quarterly and may not cover enough years. The team will collect older data where possible. However, the main forecasting model will still work without alcohol data.

The fourth risk is the same region may be named or bounded differently in different datasets. This can lead to wrong data matching. The team will work with one shared set of regions and validate the data after each merge.

The fifth is a delay in dashboard deployment. To reduce this risk, the team will create and deploy a simple version of the dashboard early on, then improve it throughout the project.

And the final risk is misuse or misinterpretation of the forecasts. If predictions are read as statements about particular communities, the system could cause harm rather than support safety. Small counts in low-population regions could also make incidents identifiable. Forecasts will therefore be shown only at region and month level, with uncertainty ranges, and small values will be suppressed.

Since Assessment 1, two additional risks have materialised during implementation. The November 2023 data gap, absent from both crime files due to the PROMIS-to-SerPro transition, was identified and retained as NaN rather than imputed, to avoid introducing artificial values into the time series. The ANZSOC category-label mismatch between the two crime files also required a manual translation table to ensure consistent offence classification across the full study period. Both issues are now resolved in the pipeline and documented in the repository change log.

9. Ethics, privacy and security considerations

Data privacy

The project only uses public and aggregated data from the Northern Territory Government and the Australian Bureau of Statistics. It does not use personal information or individual records. However, very small crime counts may still make some incidents easier to identify. Small values will therefore be hidden or grouped before being shown on the dashboard.

Indigenous data governance

Aboriginal people are strongly represented in NT crime statistics (Northern Territory Police Force, 2025), so the data must be used carefully and respectfully. The project follows the CARE Principles (Carroll et al., 2020) and the AIATSIS Code of Ethics (AIATSIS, 2020). The model will only make predictions at the regional and monthly level and will not make predictions based on Indigenous status.

Responsible use

The system is designed to help PFES plan staff and resources. It is not designed to predict whether an individual will commit a crime or to target specific addresses or communities.

Forecasts may not always be correct, so the dashboard will show uncertainty and explain the limitations of the data and model. This helps users avoid treating the predictions as exact results.

To guard against over-policing and unfair targeting, forecasts are produced at region and month level only, no suburb or address-level predictions are generated. Outputs are displayed with uncertainty ranges to make clear they are planning signals, not guaranteed outcomes. Counts below a suppression threshold are masked in the dashboard to protect individuals in low-population areas. Any operational response remains the judgment of PFES commanders rather than a direct output of the model.

Security and reproducibility

Only public data and project code will be stored in the GitHub repository. Passwords, API keys and private information will not be uploaded.

The project will also keep its data-processing and analysis steps in version-controlled code. This allows others to understand, replicate, and verify the results.

10. Evaluation of progress

All major Assessment 1 feedback points have now been addressed. The 5Vs characterisation, and leakage-free validation directly respond to the modelling feedback. The storage-architecture concern has been acknowledged with a documented justification for retaining GitHub at this milestone and a plan to revisit if concurrent access becomes necessary. Governance has been strengthened with formal working agreements covering pull request reviews, Jira completion criteria, and change logging. The ethics section has been expanded to address over-policing risks explicitly. The main items still in progress are the dashboard, deployment, and monitoring stages, which are scheduled for Weeks 8 and 10 per the workflow plan.

11. Any other relevant technical developments

Three cross-cutting developments are worth recording.

Pipeline verification. Before relying on the pipeline for modelling, it was tested end-to-end in a clean environment, surfacing several defects now fixed and recorded in the commit history:

- Two undeclared dependencies (xgboost, openpyxl) that broke the pipeline on any machine but the original developer's.
- A Windows-only Unicode crash inside a print() statement.
- A duplicated function definition from a Git merge, with a missing argument in the abandoned copy.
- A biased VIF calculation (missing its constant term) and an unsafe eval() call.

Architecture correction. The AWS-equivalent mapping (Table 2) was corrected: Athena is a query/cataloguing tool, not a transformation engine, so AWS Glue ETL replaces it as the conceptual equivalent for the cleaning/feature-engineering stage.

Output-artefact policy. The team commits pipeline outputs (eda_plots/, regression_plots/, dataset/raw/, dataset/processed/) to GitHub rather than .gitignore-excluding them, so markers can inspect results without re-running the pipeline, trading occasional merge conflicts on regenerated images for the reproducibility the module's GitHub expectations call for.

12. References

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SECTION 4 – Individual Work and Contribution Summary

1. Student Information

Name: Thi Xuan Thanh Tran (Emma)

Student ID: S389244

2. Laboratory Completion Evidence of:

Laboratory 1 – Lab: Querying Data by Using Athena (Module 4)

The screenshot shows the AWS Academy interface for the 'Querying Data by Using Athena' lab. The left sidebar contains navigation options like Account, Dashboard, Courses, Calendar, Inbox, History, and Help. The main content area shows the lab title and a list of tasks with their completion status. The right sidebar shows the total score and a list of tasks with their completion status.

Task	Status
Task 1a) Database created	5/5
Task 1b) Database table created	5/5
Task 1c) Database table previewed	5/5
Task 2) Database table jan exists	5/5
Task 4) Student queried using views	5/5
Task 5a) Student tested the query in Athena	5/5
Task 5b) CloudFormation stack created	5/5
Task 5c) CreateNamedQuery event recorded	5/5
Task 7) GetNamedQuery event recorded	5/5

3. Individual Project Contribution

Since Assessment 1, I was responsible for the data processing stage: cleaning, standardising, and merging the three source datasets (crime, population, alcohol supply) into a single analysis-ready panel. This included category translation, a two-stage region crosswalk to align police boundaries with NTG statistical regions, offence-weighted encoding of alcohol and DV flags, and population validation before aggregation. Alongside this, I contributed to team coordination through Jira task management, GitHub maintenance, report writing, and reviewing teammates' work against assignment requirements.

<https://github.com/HendrickDang/PRT661-DATA-SCIENCE-PRACTICE/commits?author=EmmaTran56>

Visualisation: https://github.com/HendrickDang/PRT661-DATA-SCIENCE-PRACTICE/commits/main/eda_plots

Documentation: <https://github.com/HendrickDang/PRT661-DATA-SCIENCE-PRACTICE/commits/main/documents>

Code: <https://github.com/HendrickDang/PRT661-DATA-SCIENCE-PRACTICE/commits/main/End%20to%20End%20pipeline.py>

4. Brief Professional Reflection

This project highlighted how much of real-world data science happens before any modelling begins. Decisions of how to handle a missing month, which geographic boundary to use, how to encode a categorical flag had direct consequences on downstream model validity. Working across both a technical role and team coordination responsibilities also developed my ability to balance execution with communication: keeping work on track through Jira, reviewing others' outputs critically, and ensuring the team's collective work met the standard the assessment required. These are skills I expect to rely on throughout my professional practice.

1. Student Information

Name: Ngoc Anh Nguyen (Will)

Student ID: S386805

2. Laboratory Completion Evidence of:

Laboratory 1 – Lab: Querying Data by Using Athena (Module 4)

The screenshot displays the AWS Academy course page for 'Querying Data by Using Athena'. The user is logged in as Will Nguyen. The main content area shows the 'Overview and objectives' section. On the right, there is a table showing the total score and a list of tasks with their completion status.

Task	Status
[Task 1a] Database created	5/5
[Task 1b] Database table created	5/5
[Task 1c] Database table previewed	5/5
[Task 2] Database table jan exists	5/5
[Task 4] Student queried using views	5/5
[Task 5a] Student tested the query in Athena	5/5
[Task 5b] CloudFormation stack created	5/5
[Task 5c] CreateNamedQuery event recorded	5/5

3. Individual Project Contribution

Since Assessment 1, I was responsible for the Ingest and Data Store stage, and for verifying that the pipeline the team built actually runs correctly end-to-end. This included implementing ingest.py to load and validate the six raw datasets with filename fuzzy-matching and manifest-based provenance tracking, compiling the project's missing dependency list after finding the pipeline would not run in a clean environment, and identifying and removing a Windows-specific Unicode crash and a duplicated function definition left over from a merge. I also corrected two silent code-quality defects - a statistically biased VIF calculation and an unsafe eval() call - before re-running the full pipeline to confirm it completes without errors ahead of the technical write-up.

Github: <https://github.com/HendrickDang/PRT661-DATA-SCIENCE-PRACTICE/commits?author=ngocanh8585>

Documentation: <https://github.com/HendrickDang/PRT661-DATA-SCIENCE-PRACTICE/commits/main/documents>

4. Brief Professional Reflection

Owning the ingest stage taught me that "does it run?" must be checked deliberately, not assumed. Defects like missing dependencies, a Windows-only Unicode crash, and a duplicated function definition were invisible on individual machines but would have surfaced during the demonstration or on the marker's machine - catching them early freed up time for modelling and reporting. It also showed me that verification means checking correctness, not just absence of errors: the VIF calculation and an unsafe eval() call ran without crashing, yet one was statistically biased and the other insecure.

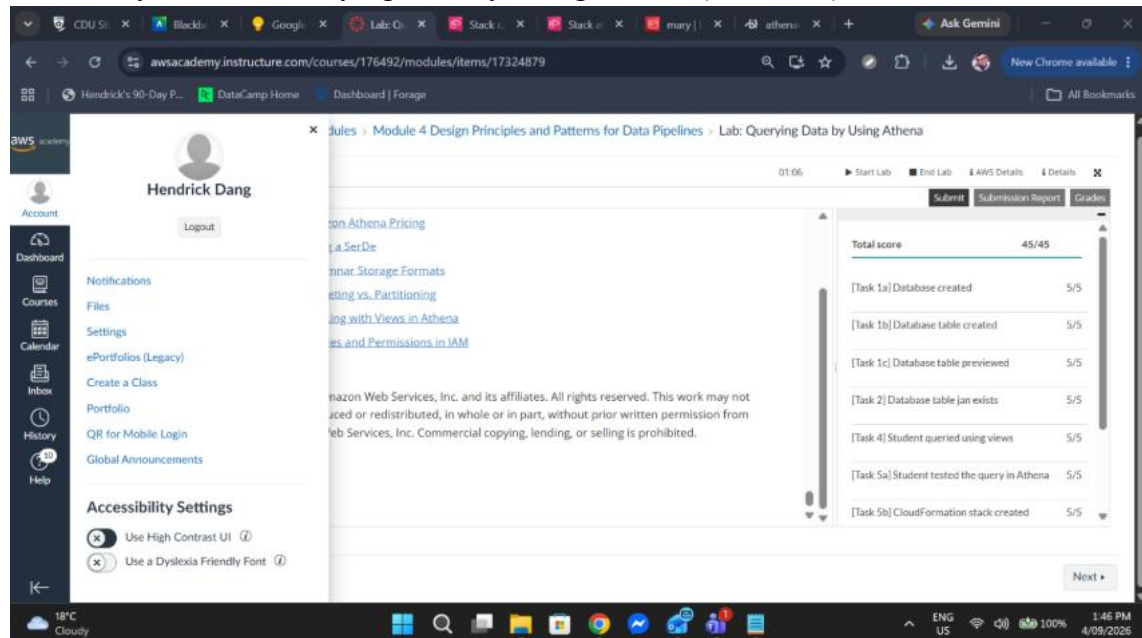
1. Student Information

Name: Van Hoi Dang (Hendrick) - Student ID: S395598

Role: Analytics and Machine Learning

2. Laboratory Completion Evidence of:

Laboratory 1 – Lab: Querying Data by Using Athena (Module 4)



3. Individual Project Contribution

Since Assessment 1 I covered the analytics and evaluation stage, addressing the modelling feedback. I built `evaluation.py`, a standalone module providing naive baselines, leakage-free feature selection, rolling-origin backtesting at four horizons, prediction intervals and structural-break indicators. It exposed two leakage sources I then fixed: variant selection on the test set, and scaling fitted before the CV split. Seasonal naive and 12-month moving average benchmarks produced Table 13 (Section 4.6). I also defined the forecast specification, drift thresholds and retraining rules, and wrote the Forecast Evaluation and Benchmarking subsection (Section 4.6).

Github: <https://github.com/HendrickDang/PRT661-DATA-SCIENCE-PRACTICE/commits?author=HendrickDang>

Documentation: <https://github.com/HendrickDang/PRT661-DATA-SCIENCE-PRACTICE/commits/main/documents>

Evaluation: <https://github.com/HendrickDang/PRT661-DATA-SCIENCE-PRACTICE/commits/main/evaluation.py>

Backtest: <https://github.com/HendrickDang/PRT661-DATA-SCIENCE-PRACTICE/commits/main/Outputs>

4. Brief Professional Reflection

A model comparison only means something against a benchmark. Assessment 1 compared regression models against each other; once naive baselines were added, a 12-month moving average beat every model from three months out, and XGBoost only won at one month. Reporting a split result was less comfortable than reporting a single winner, but more defensible.

The second lesson was leakage: every case I found produced plausible output, which is why they were easy to miss.

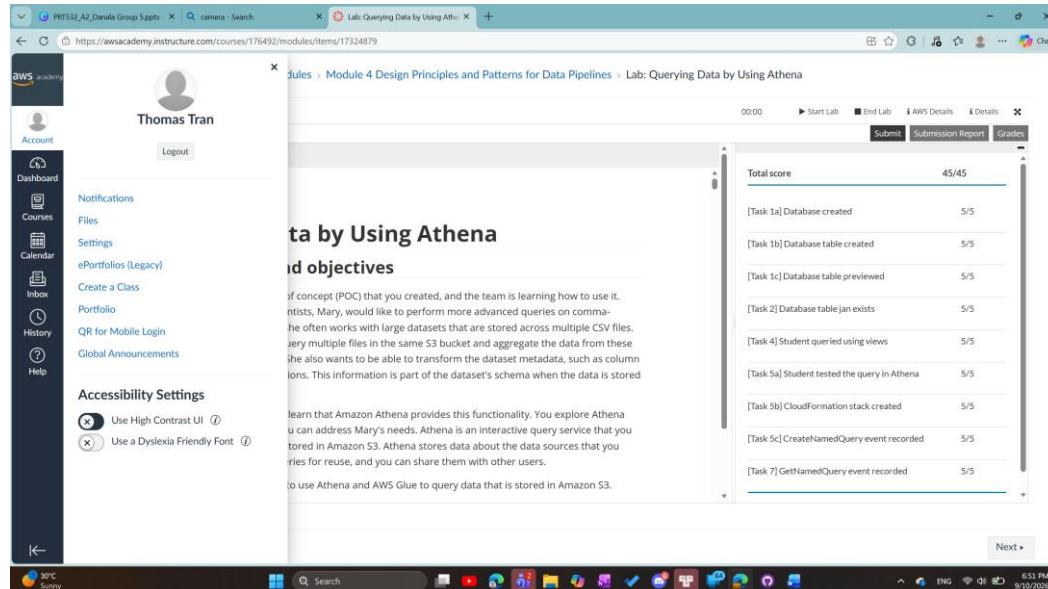
1. Student Information

Name: Le Nhat Minh Tran (Thomas)

Student ID: S390789

2. Laboratory Completion Evidence of:

Laboratory 1 – Lab: Querying Data by Using Athena (Module 4)



3. Individual Project Contribution

In this assessment, I maintain multiple roles, mostly with the technical side of the project. My contribution includes: Review the output from the ingesting and preprocessing steps, doing basic testing to see if anomalies exist and also seeking clarification when necessary, making sure that the processed data are ready for the next steps; Researching and choosing appropriate machine learning models and techniques from wide range of options and implementing them in the pipeline; Documenting and writing up reports based on my findings.

GitHub commits: [Commits · HendrickDang/PRT661-DATA-SCIENCE-PRACTICE](#)

Documentation: [PRT661-DATA-SCIENCE-PRACTICE/documents](#) at [main](#) · [HendrickDang/PRT661-DATA-SCIENCE-PRACTICE](#)

4. Brief Professional Reflection

Throughout this assessment, I focused mainly on the technical components of the project, which strengthened my analytical, research, and problem-solving skills. Working with different machine learning models improved my understanding of model selection, evaluation, and optimisation, giving me practical experience applying theory to a real-world problem. Conducting research to support our findings also exposed me to important concepts and current developments in machine learning. This broadened my technical knowledge and deepened my appreciation for the academic foundations that guide data science and predictive analytics. A major lesson from the project was the importance of collaboration and communication. Regular discussions with teammates and tutors helped me understand how professional IT projects are coordinated. Sharing updates, seeking feedback, and resolving issues highlighted the value of teamwork, stakeholder engagement, and clear communication. Overall, the project strengthened both my technical capabilities and professional skills. It reinforced the importance of data quality, evidence-based decision-making, continuous learning, and effective collaboration when developing reliable and impactful data-driven solutions.